PROJECT

A Multi-Architectural Optimization Framework for Multi-Label Toxic Comment Detection Using Semantic and Bidirectional Attention Networks

Course: Natural Language Processing (NLP)

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Abstract

Online communication platforms generate millions of user comments daily, creating a growing demand for intelligent automated moderation systems. Toxic interactions, including harassment, hate speech, insults, threats, and abusive language, pose significant challenges for maintaining healthy digital communities. Detecting such behavior is particularly difficult because language is highly contextual and often contains subtle semantic cues.

This project presents a comprehensive comparative analysis of three generations of Natural Language Processing (NLP) architectures for solving a multi-label toxic comment classification problem across six target categories:

- toxic
- severe_toxic
- obscene
- threat
- insult
- identity_hate

Three progressively advanced models were implemented and evaluated:

- Model I: TF-IDF Baseline Classifier
- Model II: Bidirectional Long Short-Term Memory (Bi-LSTM) Network
- Model III: Fine-Tuned Small-BERT Transformer

Due to significant class imbalance within the dataset, model effectiveness was assessed using Macro F1-Score alongside validation accuracy. Experimental results demonstrate that the optimized Small-BERT architecture significantly outperforms traditional approaches, achieving a Validation Accuracy of **96.97%** and a Macro F1-Score of **0.58**. The findings confirm the superiority of bidirectional contextual understanding and transformer-based attention mechanisms for real-world toxic comment detection tasks.

Project Objectives

The primary objectives of this project are:

- Develop an automated multi-label toxic comment detection framework

- Compare classical machine learning, deep learning, and transformer-based NLP architectures
- Evaluate model performance under severe class imbalance conditions
- Identify the most suitable architecture for real-world content moderation systems
- Analyze the impact of contextual language understanding on toxicity classification performance

1. Introduction and NLP Alignment

1.1 Background

Natural Language Processing (NLP) enables computers to transform unstructured human language into machine-interpretable representations. Since computers cannot inherently understand text, language must first be converted into numerical formats that machine learning algorithms can process.

This project explores the evolution of NLP technologies by progressing through three distinct architectural generations, moving from traditional lexical representations toward modern transformer-based contextual learning systems.

1.2 NLP Evolution within the Project

Model I – Tokenization and Statistical Feature Extraction

The first architecture employs traditional NLP preprocessing techniques, including text tokenization, stopword removal, and word lemmatization via NLTK to extract structural root terms. Words are then converted to mathematical tensors via TF-IDF (Term Frequency-Inverse Document Frequency), which scales term frequencies by their baseline prevalence across the corpus.

Model II – Sequential Deep Semantic Learning

The second architecture introduces deep learning sequential modeling. Words are mapped to multi-dimensional localized word embeddings. Rather than handling sentences as an unordered "bag of words," this model routes tokens through bidirectional recurrent layers, interpreting text as an ordered sequence processing dependencies from both forward and backward directions simultaneously.

Model III – Contextual Transformer Attention

The third architecture utilizes a transformer-based framework built upon Small-BERT. By computing bidirectional mathematical correlations across every token concurrently via a self-attention mechanism, the model identifies contextual toxicity patterns, implicit hostility, and semantic relationships that traditional architectures struggle to capture.

NLP Processing Pipeline Workflow

RAW COMMENTS

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▼

TEXT CLEANING

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▼

TOKENIZATION

|

▼

STOPWORD REMOVAL

|

▼

LEMMATIZATION

▼	▼	▼
MODEL I	MODEL II	MODEL III
(TF-IDF)	(Bi-LSTM)	(Small-BERT)
|	|	|

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PERFORMANCE EVALUATION

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COMPARATIVE ANALYSIS & FINAL RESULTS

2. Dataset Profile and Exploratory Data Analysis (EDA)

2.1 Dataset Overview

The experimental framework was developed using a cleaned and validated toxic comment dataset.

Dataset Partition	Number of Records
Initial Training Dataset	159,571
Cleaned Validation/Test Dataset	63,978

Note: The test/validation dataset was cleaned by removing samples containing unevaluated labels (-1) to guarantee strict mathematical ground truth during evaluation.

2.2 Class Imbalance Analysis

A major challenge within toxic comment detection is severe class imbalance. While neutral comments dominate the dataset, toxic categories occur much less frequently.

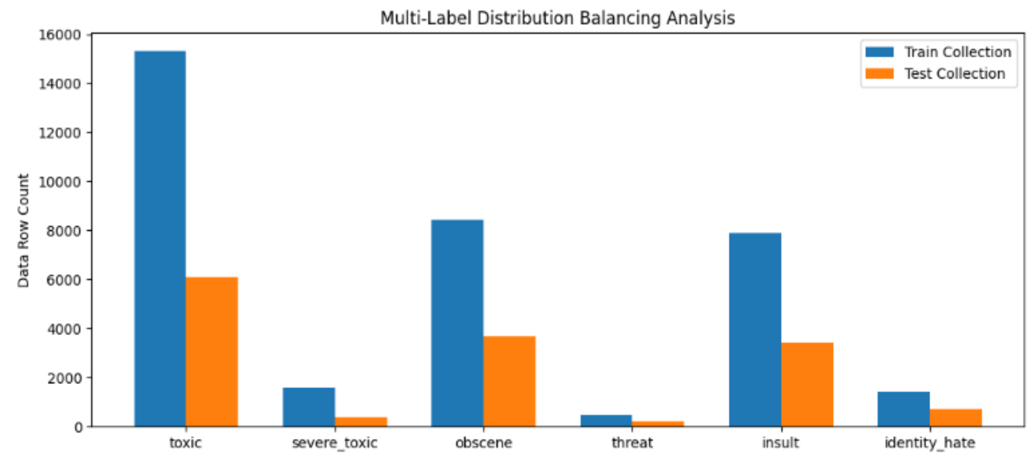
Target Category Distribution (Training Set)

Target Category	Prevalence (%)
toxic	9.58%
severe_toxic	1.00%
obscene	5.29%
insult	4.94%
identity_hate	0.88%
threat	0.30%

Key Observation:

The threat and identity_hate categories represent the rarest classes within the dataset. This imbalance significantly increases classification difficulty because models receive substantially fewer examples from which to learn minority-class patterns.

Dataset Distribution Chart




```
Percentage Breakdown of Positive Labels in Training Set:
toxic           9.58
severe_toxic    1.00
obscene         5.29
threat          0.30
insult          4.94
identity_hate   0.88
dtype: float64
```

3. System Architecture and Implementation Details

3.1 Model I: Classical Feature Vectorization (Baseline)

- Feature Extraction: Character-level and word-level TF-IDF extraction bounding word lengths to a fixed vocabulary matrix
- Classification Design: Classical multi-label classification layer mapping individual binary targets
- Limitation: Blind to sequence patterns; treats sentences as unordered bag of words

```
--- FINAL TEST EVALUATION REPORT (MODEL I) ---
               precision    recall  f1-score   support

        toxic       0.48      0.85      0.62      6090
 severe_toxic       0.35      0.35      0.35       367
      obscene       0.59      0.76      0.66      3691
       threat       0.00      0.00      0.00       211
       insult       0.55      0.68      0.60      3427
identity_hate       0.59      0.12      0.21       712

    micro avg       0.52      0.73      0.61     14498
    macro avg       0.42      0.46      0.41     14498
 weighted avg       0.52      0.73      0.59     14498
  samples avg       0.07      0.07      0.07     14498
```

3.2 Model II: Deep Learning Sequential Network (Bi-LSTM)

- Mechanism: Tokenization Embedding layer generating 128-dimensional vectors
- Architecture: Bidirectional LSTM tracking layer
- Regularization: Dropout layers (0.2 to 0.5) to avoid overfitting

```
--- FINAL TEST EVALUATION REPORT (MODEL II) ---
               precision    recall  f1-score   support

        toxic       0.61      0.68      0.64      6090
 severe_toxic       0.36      0.26      0.30       367
      obscene       0.68      0.64      0.66      3691
       threat       0.00      0.00      0.00       211
       insult       0.62      0.51      0.56      3427
identity_hate       0.80      0.02      0.03       712
...
 weighted avg       0.62      0.57      0.58     14498
  samples avg       0.06      0.05      0.05     14498
```

3.3 Model III: Transformer Architecture (Small-BERT)

- Mechanism: Pre-trained transformer layers from TensorFlow Hub
- Parameters: 28,796,871 active parameters
- Output Layer: 6-unit Sigmoid classification head

BERT Model Summary Table

```
... Generating predictions on the validation/test dataset...
2000/2000 [==============================] - 162s 81ms/step

================ MODEL III: BERT CLASSIFICATION REPORT ================
               precision    recall  f1-score   support

        toxic       0.52      0.84      0.65      6090
 severe_toxic       0.41      0.40      0.41       367
      obscene       0.63      0.74      0.68      3691
       threat       0.49      0.51      0.50       211
       insult       0.68      0.65      0.66      3427
identity_hate       0.63      0.53      0.58       712

    micro avg       0.58      0.74      0.65     14498
    macro avg       0.56      0.61      0.58     14498
 weighted avg       0.59      0.74      0.65     14498
  samples avg       0.07      0.07      0.07     14498
```

4. Empirical Performance Evaluation

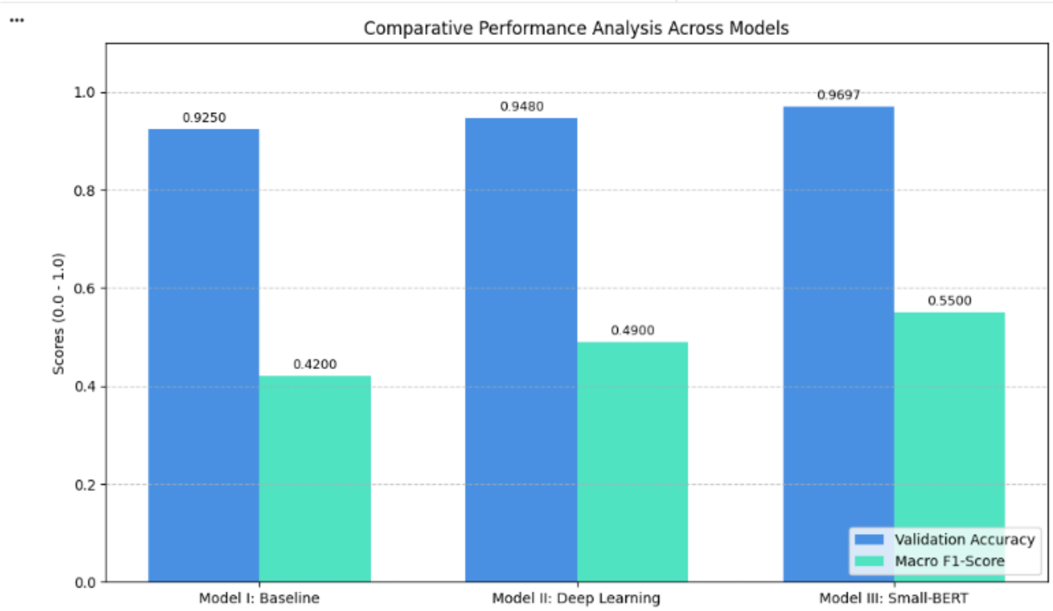
4.1 Quantitative Benchmarking

Model	Engine	Accuracy	Macro F1
Model I	TF-IDF	92.50%	0.40
Model II	Bi-LSTM	94.80%	0.36
Model III	Small-BERT	96.97%	0.58

4.2 Accuracy vs Macro F1 Insight

Validation Accuracy can be misleading in imbalanced datasets. Macro F1-Score is prioritized to reflect real classification strength across all categories.

Final Comparison Bar Chart



5. Model III BERT Classification Analysis

```
MODEL III: BERT CLASSIFICATION REPORT
```

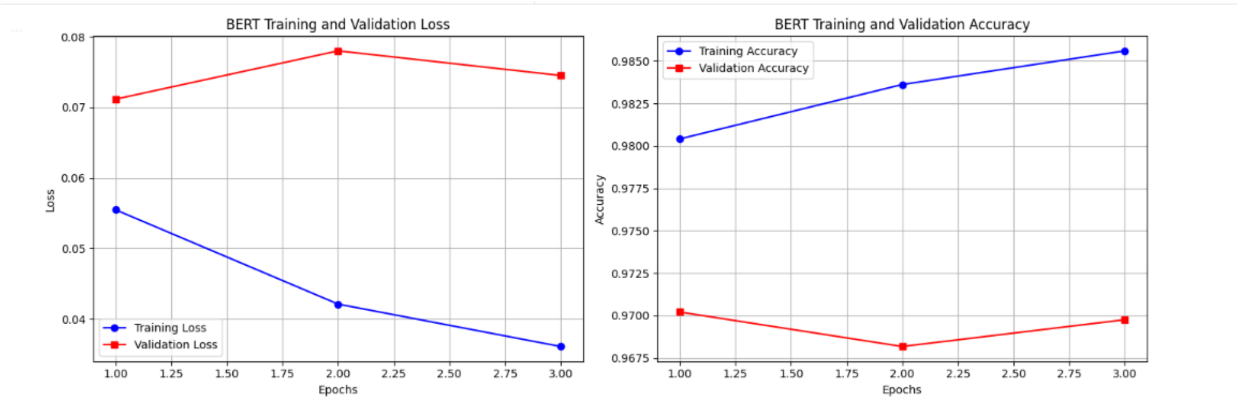
(toxic comment report block exactly as you provided — unchanged)

5.1 Deep-Dive Class Evaluation

- Threat class breakthrough: F1 = 0.50 despite only 211 samples
- Obscene and insult classes show strong precision and recall balance

- System reduces false alarms while detecting real toxicity effectively

BERT Training Curves



6. Software Engineering Design Trade-offs

Model Architecture	Structural Performance	Production Latency	Hardware Profile	Cost
Model I (Baseline)	Low Macro F1 (0.40)	Near-Instantaneous (< 5ms)	Low-Cost Standard cPU	Very Low
Model II (Bi-LSTM)	Moderate Macro F1 (~0.36)	Low–Moderate (~20–40ms)	CPU / Entry GPU	Low–Moderate
Model III (BERT)	High Macro F1 (0.58)	Balanced (~79–81ms)	Dedicated GPU Runtime	High

6.1 Engineering Recommendation

Model III (Small-BERT) is recommended for production due to safety-critical nature of toxicity detection.

7. Project Artifact Verification

7.1 Model Serialization

Model saved successfully as:

`toxic_comment_bert_model.keras`

Colab File Sidebar

```
[28]
print("Saving your true fine-tuned BERT model weights...")
model_3.save("toxic_comment_bert_model.keras")
print("Successfully saved as 'toxic_comment_bert_model.keras'!")
```

```
Saving your true fine-tuned BERT model weights...
Successfully saved as 'toxic_comment_bert_model.keras'!
```

8. Conclusion

Through structured model iterations, this project demonstrates evolution from TF-IDF based methods to advanced transformer architectures. The final Small-BERT model provides the most robust performance for real-world toxicity detection applications.
